However, as we shall now see, neither of these two strategies, hawk or dove, would in fact be evolutionarily stable on its own, and we should therefore not expect either of them to evolve. To show this we must calculate average pay-offs.

Suppose we have a population consisting entirely of doves. Whenever they fight, nobody gets hurt. The contests consist of prolonged ritual tournaments, staring matches perhaps, which end only when one rival backs down. The winner then scores 50 points for gaining the resource in dispute, but he pays a penalty of −10 for wasting time over a long staring match, so scores 40 in all. The loser also is penalized −10 points for wasting time. On average, any one individual dove can expect to win half his contests and lose half. Therefore his average pay-off per contest is the average of +40 and −10, which is +15. Therefore, every individual dove in a population of doves seems to be doing quite nicely.

But now suppose a mutant hawk arises in the population. Since he is the only hawk around, every fight he has is against a dove. Hawks always beat doves, so he scores +50 every fight, and this is his average pay-off. He enjoys an enormous advantage over the doves, whose net pay-off is only +15. Hawk genes will rapidly spread through the population as a result. But now each hawk can no longer count on every rival he meets being a dove. To take an extreme example, if the hawk gene spread so successfully that the entire population came to consist of hawks, all fights would now be hawk fights. Things are now very different. When hawk meets hawk, one of them is seriously injured, scoring −100, while the winner scores +50. Each hawk in a population of hawks can expect to win half his fights and lose half his fights. His average expected pay-off per fight is therefore half-way between +50 and −100, which is −25. Now consider a single dove in a population of hawks. To be sure, he loses all his fights, but on the other hand he never gets hurt. His average pay-off is 0 in a population of hawks, whereas the average pay-off for a hawk in a population of hawks is −25. Dove genes will therefore tend to spread through the population.

The way I have told the story it looks as if there will be a continuous oscillation in the population. Hawk genes will sweep to ascendancy; then, as a consequence of the hawk majority, dove genes will gain an advantage and increase in numbers until once again hawk genes start to prosper, and so on. However, it need not be an oscillation like this. There is a stable ratio of hawks to doves. For the particular arbitrary points